

Simulation-Based Inference

Exercise 7

1. ABC-MCMC for a Gaussian mean

Given the model, $\mu \sim \mathcal{N}(0, 1)$ $y_i \mid \mu \sim \mathcal{N}(\mu, 1)$, $i = 1, \dots, N$.

The **sufficient summary statistic** for μ is the sample mean $T(y) = \bar{y}$, so the distance is simply $|\bar{y}^{(p)} - \bar{y}|$.

The **analytic posterior** is $\mu \mid \tilde{y} \sim \mathcal{N}\left(\frac{N\tilde{y}}{N+1}, \frac{1}{N+1}\right)$.

- Generate 5 observations $\tilde{y} \sim \mathcal{N}(\mu = 1, 1)$, run ABC-MCMC using any programming language of your choice, and overlay the analytic posterior.
- Repeat the above for 50 observations, and compare the results to the analytic posterior.
- Plot the trace of the chain from **1b** and report the fraction of steps where a new proposal was accepted (acceptance rate).

2. ABC with summary statistics

Use the same model as in Question 1 but now with the following summary statistics:

Summary statistic $T(y)$	Sufficient?
sample mean $\bar{y}$	Yes
sample median	No
first observation y_1 only	No

- Using a programming language of your choice, generate $N = 20$ observations from $\mathcal{N}(\mu = 1, 1)$. Run ABC rejection sampling three times — once per summary statistic — using the same number of prior draws and the same ϵ for all three. Plot the resulting ABC posteriors alongside the analytic posterior and compare.
- Using only summary statistic sample mean, vary $\epsilon \in \{0.5, 0.15, 0.05\}$ and overlay the resulting ABC posteriors on a single plot together with the analytic posterior. Describe how the approximation changes with ϵ .